Open-System Decoherence and Holographic Phase Dynamics

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ABSTRACT

We derive a dimensionless, holographically bounded parameterization of open-system quantum decoherence along curved spacetime worldlines. Combining the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation with Caldeira–Leggett thermal photon scattering and Penrose–Diósi gravitational self-energy dephasing, we construct the cumulative **Decoherence Action** (Ω). To prevent unphysical vanishing of quantum state purity near causal horizons, we impose the Bekenstein–Hawking holographic Hilbert-space dimensionality bound to establish an irreducible **Holographic Coherence Floor** (Φ_{floor}) and define **Timeline Integrity** (Φ_{τ}). Finally, coupling Φ_{τ} to the Margolus–Levitin orthogonalization rate yields the count of **Surviving Coherent Cycles** ($\mathcal{N}_{eff}$) and the dimensionless **Lloyd–Bekenstein Gate Ratio** (η_{Lloyd}).

1. First-Principles Mathematical Derivations

1.1 Open Quantum Evolution and Decoherence Action (Ω)

Consider a quantum system of characteristic radius L and effective mass M_{eff} interacting with an ambient thermal and gravitational environment along its proper worldline (τ_{pre}). Under the Born–Markov approximation, the reduced density matrix $\rho_S(\tau_{pre}) = \mathrm{Tr}_E[\rho_{tot}]$ evolves according to the **Gorini–Kossakowski–Sudarshan–Lindblad (GKSL)** master equation:

$$d\rho_S / d\tau_{pre} = -(i/\hbar)[H_S, \rho_S] + \Sigma_k \Gamma_k (L_k \rho_S L_k^\dagger - ½\{L_k^\dagger L_k, \rho_S\})$$

For a spatial or energy superposition between two orthogonal states $|1\rangle$ and $|2\rangle$, off-diagonal coherence decays exponentially with the integrated environmental scattering rate:

$$|\rho_{12}(\tau_{pre})| = |\rho_{12}(0)| \cdot \exp(-\int_0^{t_{pre}} \Gamma_{tot}(\tau)\, d\tau)$$

The total dephasing rate $\Gamma_{tot} = \Gamma_{th} + \Gamma_{grav}$ comprises two competing physical channels:

1. **Caldeira–Leggett Thermal Scattering (Γ_{th}):** Scattering of blackbody quanta at local effective temperature T_{eff} (with reduced thermal wavelength $\lambda\!\!\!^{-}_{th} = \hbar c / (k_B T_{eff})$) across cross-sectional area $A \sim L^2$ yields:

$$\Gamma_{th} = [\, L^2\, (k_B\, T_{eff})^3 \,] \,/\, [\, (\hbar c)^2\, \hbar \,] = (c \cdot L^2) \,/\, \lambda\!\!\!^{-}_{th}{}^3$$

2. **Penrose–Diósi Gravitational Self-Energy Rate (Γ_{grav}):** A mass distribution M_{eff} displaced across scale L carries gravitational self-energy uncertainty $\Delta E_{grav} \sim G\, M_{eff}^2 / L = (M_{eff} / M_P)^2\, (\hbar c / L)$, giving the gravitational dephasing rate:

$$\Gamma_{grav} = \Delta E_{grav} \,/\, \hbar = (M_{eff} \,/\, M_P)^2\, (c \,/\, L)$$

Integrating Γ_{tot} over elapsed proper time $t_{pre} = t_{obs} / \rho_\tau$ defines the dimensionless **Decoherence Action (Ω)**:

$$\Omega \equiv \int_0^{t_{pre}} (\Gamma_{th} + \Gamma_{grav})\, d\tau = (\Gamma_{th} + \Gamma_{grav}) \cdot t_{pre}$$

1.2 Holographic Hilbert-Space Dimension and Residual Coherence Floor (Φ_{floor})

In naive infinite-dimensional quantum mechanics, $\lim_{\Omega \to \infty} e^{-\Omega} = 0$, implying complete erasure of quantum state purity. However, within a causal spherical boundary of radius L, the **Bekenstein–Hawking holographic entropy bound** ($S_{max} / k_B = A / 4\ell_P^2$) restricts the physical Hilbert space to a finite effective dimension N_{dof}.

Let $\zeta = \log_{10}(L / \ell_P)$ denote the logarithmic radius in Planck units ($L / \ell_P = 10^\zeta$). The bounding surface area in Planck units is $A / \ell_P^2 = 4\pi (L / \ell_P)^2 = 4\pi \cdot 10^{2\zeta}$, yielding the effective number of orthogonal boundary microstates:

$$N_{dof} = 1 + A \,/\, (4\ell_P^2) = 1 + \pi \cdot 10^{2\zeta}$$

When an initially pure quantum state ($\mathrm{Tr}(\rho_S^2) = 1$) completely thermalizes into a maximally mixed state over N_{dof} accessible microstates ($\rho_{mix} = I / N_{dof}$), its purity reaches the irreducible inverse-dimension **Holographic Coherence Floor (Φ_{floor})**:

$$\Phi_{floor} \equiv \mathrm{Tr}(\rho_{mix}^2) = 1 \,/\, N_{dof} = 1 \,/\, (1 + \pi \cdot 10^{2\zeta})$$

At the Planck scale ($L = \ell_P$, $\zeta = 0$), $\Phi_{floor} = (1 + \pi)^{-1} \approx 0.2414$; at macroscopic laboratory scales ($L = 1$ m, $\zeta \approx 34.79$), $\Phi_{floor} \sim 10^{-70}$, smoothly recovering classicality while strictly preserving unitarity.

1.3 Timeline Integrity (Φ_τ)

Interpolating smoothly between initial pure coherence ($\Phi_\tau(0) = 1$) and maximally mixed holographic equilibrium ($\Phi_\tau(\infty) = \Phi_{\text{floor}}$) under GKSL exponential damping defines **Timeline Integrity (Φ_τ)**:

$$\Phi_\tau(\Omega) = \Phi_{\text{floor}} + (1 - \Phi_{\text{floor}}) \cdot e^{-\Omega}$$

This function satisfies $\Phi_\tau(0) = 1$, $\lim_{\Omega \to \infty} \Phi_\tau(\Omega) = \Phi_{\text{floor}} > 0$, and strict monotonicity $d\Phi_\tau / d\Omega = -(1 - \Phi_{\text{floor}})e^{-\Omega} \leq 0$ for all $L \geq \ell_P$.

1.4 Surviving Coherent Cycles ($\mathcal{N}_{\text{eff}}$) and Lloyd–Bekenstein Gate Ratio (η_{Lloyd})

By the **Margolus–Levitin quantum speed limit**, a quantum system with local energy $E = h\, f_{\text{local}}$ requires a minimum orthogonalization interval $\Delta\tau_\perp = h / (4E) = (4f_{\text{local}})^{-1}$, accumulating $\mathcal{N}_\varphi = f_{\text{local}} \cdot t_{\text{pre}}$ unscattered phase cycles over proper duration t_{pre}. Weighting raw cycles by Φ_τ yields the **Surviving Coherent Cycles ($\mathcal{N}_{\text{eff}}$)**:

$$\mathcal{N}_{\text{eff}} \equiv \mathcal{N}_\varphi \cdot \Phi_\tau = (f_{\text{local}} \cdot t_{\text{pre}}) \cdot [\, \Phi_{\text{floor}} + (1 - \Phi_{\text{floor}})\, e^{-\Omega}\,]$$

Comparing $\mathcal{N}_{\text{eff}}$ against the system's maximum holographic information capacity ($S_{\text{BH}} / k_B = A / 4\ell_P^2$) defines the dimensionless **Lloyd–Bekenstein Gate Ratio (η_{Lloyd})**:

$$\eta_{\text{Lloyd}} \equiv \mathcal{N}_{\text{eff}} \,/\, (S_{\text{BH}} \,/\, k_B)$$

2. Plain-Language Physical and Operational Interpretation

- **Decoherence Action (Ω):** Acts as a dimensionless environmental scattering counter. When $\Omega \ll 1$, the quantum wavepacket evolves without significant environmental disturbance; once $\Omega > 1$, phase-randomizing collisions accumulate rapidly, driving the transition toward classical behavior.
- **Holographic Coherence Floor (Φ_{floor}):** Ensures that quantum information never drops to absolute zero purity. Even inside a thermalized black hole horizon, Φ_{floor} preserves the exact minimum microstate correlation ($1 / N_{\text{dof}}$) required by unitary holographic physics.
- **Timeline Integrity (Φ_τ):** Measures the surviving fraction of a system's initial quantum superposition. It starts at 1.0000 (100% pure coherence) and decays exponentially toward Φ_{floor} as Ω increases.
- **Surviving Coherent Cycles ($\mathcal{N}_{\text{eff}}$):** Counts how many usable, phase-coherent quantum oscillations occur before environmental noise washes out interference. Once Φ_τ saturates at Φ_{floor}, coherent cycle growth halts.
- **Lloyd–Bekenstein Gate Ratio (η_{Lloyd}):** Benchmarks coherent dynamical evolution against holographic bit capacity. Values $\eta_{\text{Lloyd}} \leq 1$ confirm adherence to black hole information bounds, whereas $\eta_{\text{Lloyd}} > 1$ identifies delocalized elementary particles in flat space operating far outside their Schwarzschild radius.

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